Final Project Report

KnowDocs: Chat with Technology Documents (RAG)

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 202c

Problem Statement & Objectives

- **Problem Statement:** Architect a complete Retrieval-Augmented Generation (RAG) system for Technology document intelligence. The solution enables users to upload professional documents (PDF reports, manuals) which are then processed through a pipeline of text chunking and vector embedding. By leveraging a high-performance Vector Database and LLM APIs, the system facilitates accurate, context-aware Q&A sessions, ensuring that responses are grounded strictly in the provided domain-specific content.

- **Project Objectives:** To develop a "Retrieval-Augmented Generation" (RAG) chatbot system that provides accurate, grounded answers to user queries by fetching relevant information from an internal document knowledge base.

- **Scope of the Project:** The project covers document ingestion (PDF/HTML), semantic indexing via embeddings, and a runtime query-answering interface.

2. Proposed Solution

KnowDocs is a system that uses an LLM augmented by a vector database to ensure responses are based on actual document content.

Key Features:

- **Document Ingestion:** Parsing and extracting text from PDF and HTML files.
- **Semantic Search:** Using vector embeddings to find the most relevant document passages.
- **Grounded Generation:** Inserting retrieved context into LLM prompts to reduce hallucinations.
- **Web Interface:** A user-friendly chat UI for interacting with the system.

3. Overall Architecture / Workflow

The system utilizes a three-layer architecture consisting of a **Streamlit Frontend**, **Python/FastAPI Backend**, and a **FAISS Data Layer**.

- **Ingestion Pipeline:** Documents are parsed, split into overlapping chunks, converted into embeddings using a Sentence-Transformer model, and stored in a FAISS index.
- **Query Pipeline:** User questions are embedded to perform a similarity search in FAISS. The top relevant chunks are used as context for the LLM (Groq API) to generate a response.

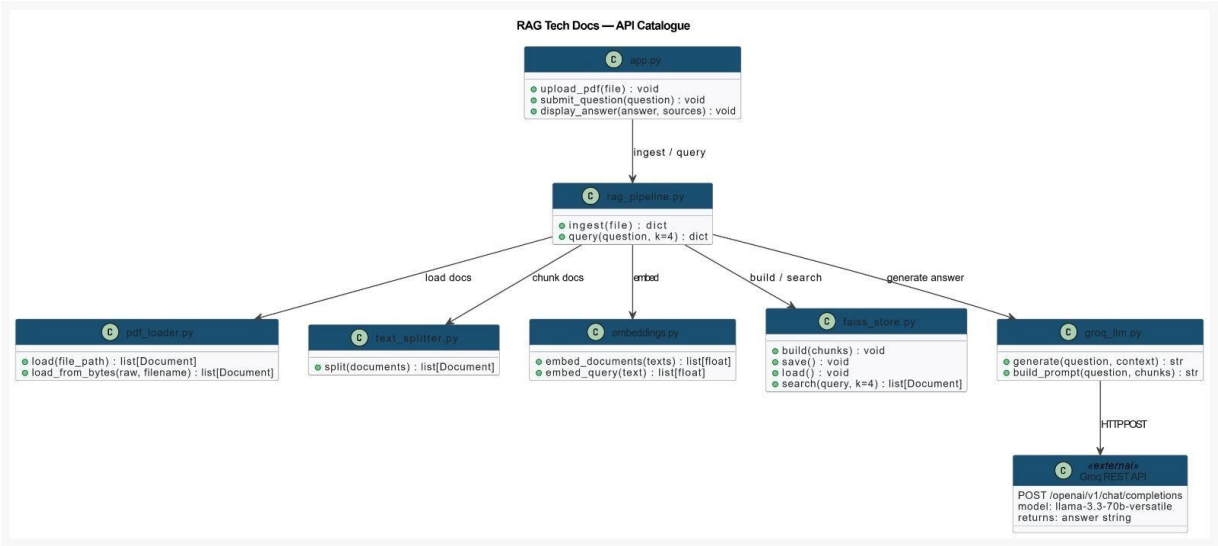

Figure 1: API Catalogue

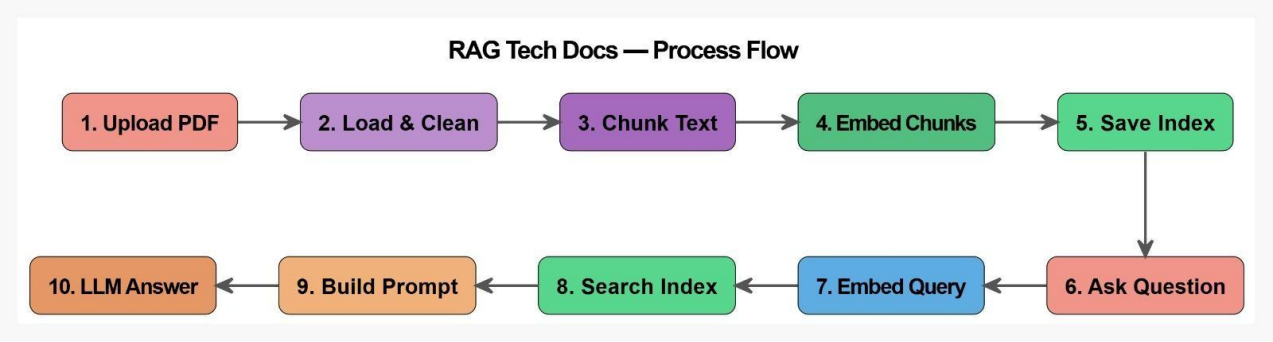

Figure 2: Process Flow

4. Tools & Technologies Used

- **Frontend:** Streamlit
- **Backend:** Python, FastAPI
- **LLM API:** Groq (Llama-3.3-70b-versatile)
- **Vector Database:** FAISS (Facebook AI Similarity Search)
- **Embedding Model:** all-MiniLM-L6-v2
- **Document Processing:** PyPDF / pdfminer
- **Containerization:** Docker Compose

5. Results & Output

- **Key Outcomes:** The system successfully mitigates LLM hallucinations by grounding answers in retrieved facts. It provides a fast, memory-optimized search experience using FAISS.
- **Performance:** Achieves low-latency retrieval via in-memory FAISS searches, with the primary delay being the external LLM API call.

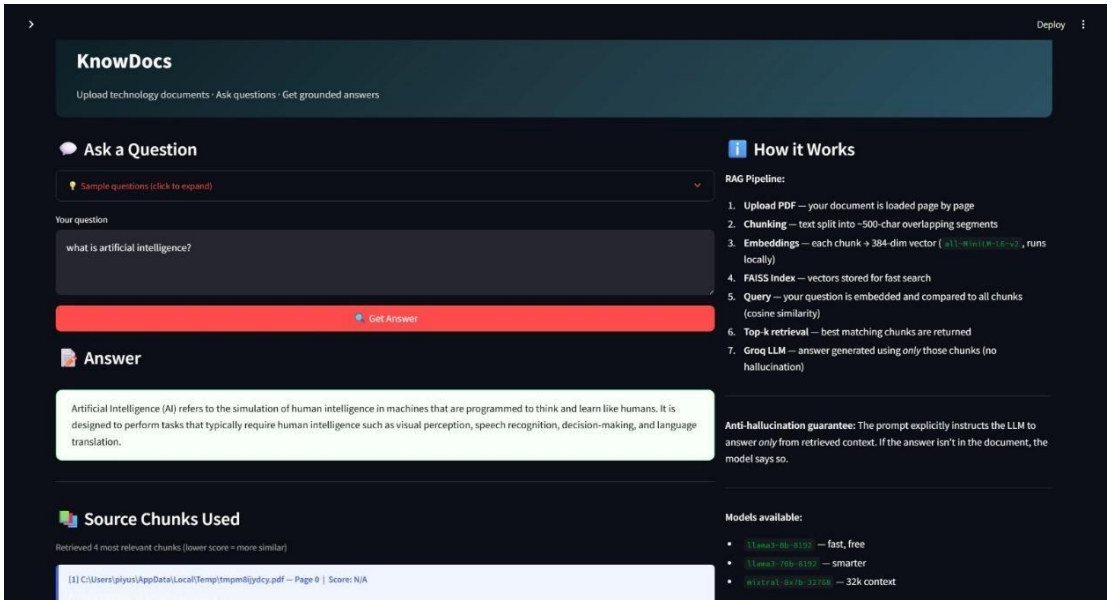

Figure 3: Screenshot 1

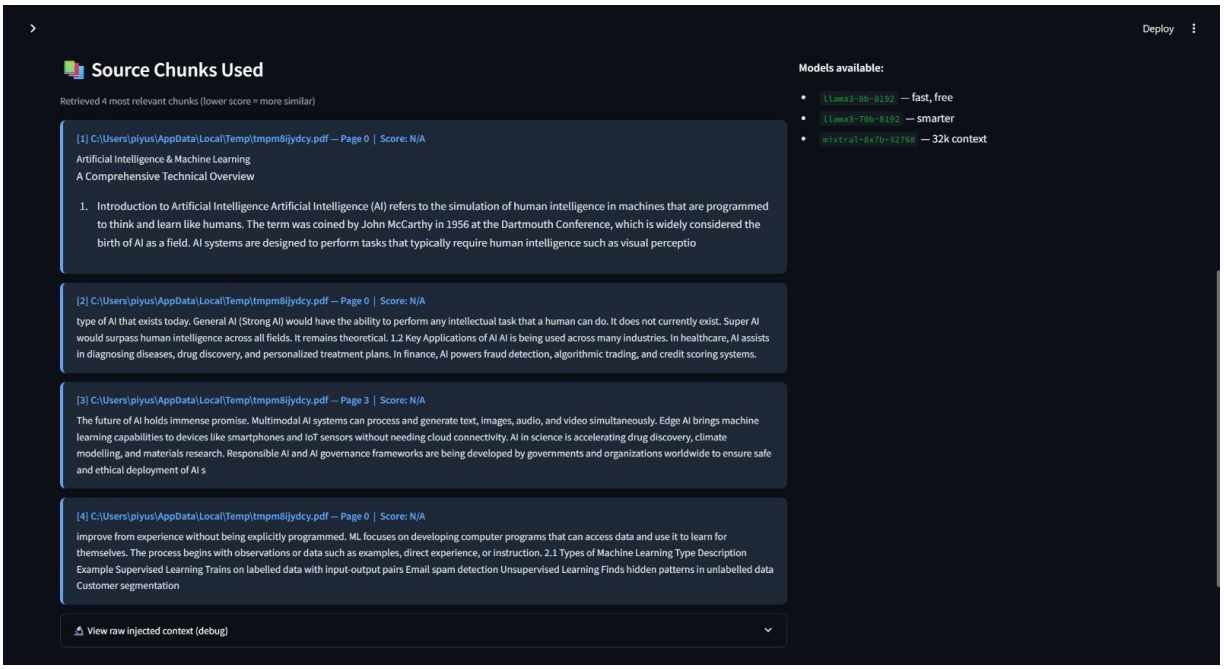

Figure 4: Screenshot 2

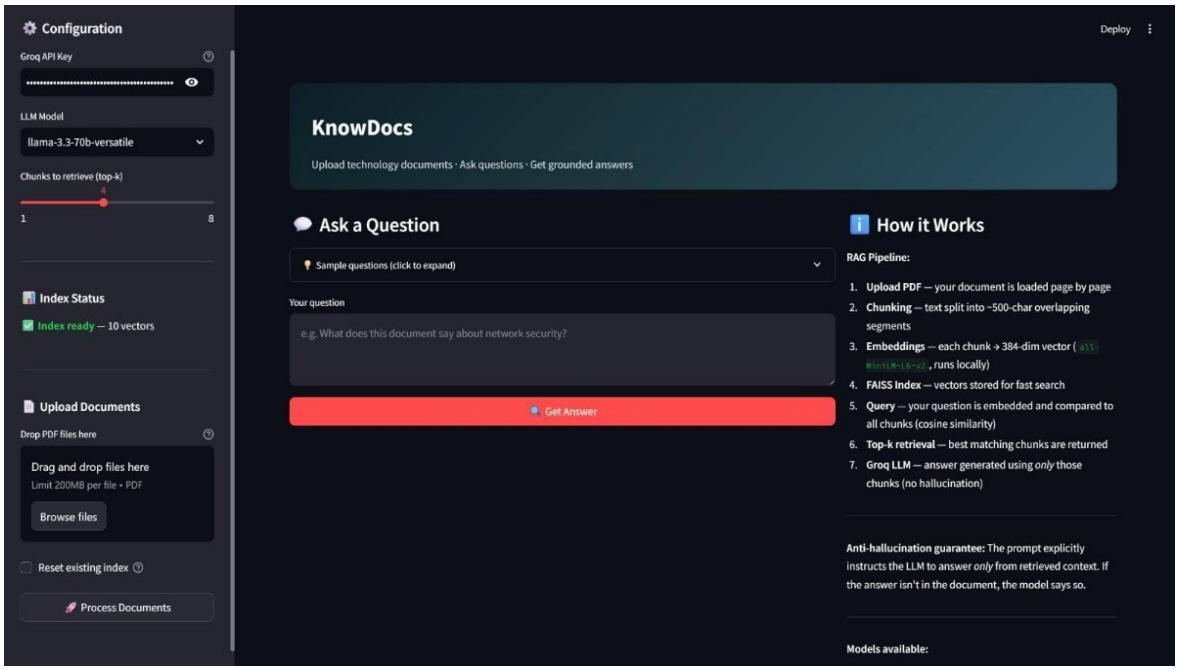

Figure 5: Screenshot 3

6. Conclusion

KnowDocs demonstrates an effective implementation of the RAG framework for technical documentation. By integrating semantic search with high-performance LLMs like Llama-3.3 via Groq, the project achieves a system that is both accurate and scalable. Key learnings include the importance of effective text chunking and the necessity of re-indexing if embedding models are updated.

7. Future Scope & Enhancements

- **Scaling:** Implementation of GPU acceleration or sharding for the FAISS index to handle even larger document sets.
- **Optimization:** Further caching of frequent query embeddings and results to reduce API costs and improve response times.
- **Advanced UI:** Adding more granular metadata tracking to allow users to link answers directly back to specific page numbers or source IDs